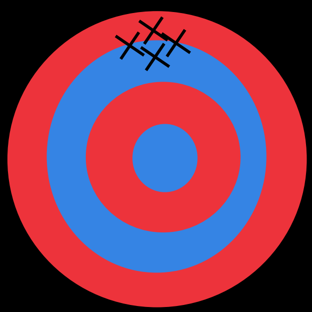
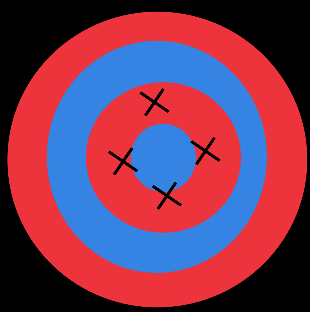
Accuracy vs Precision

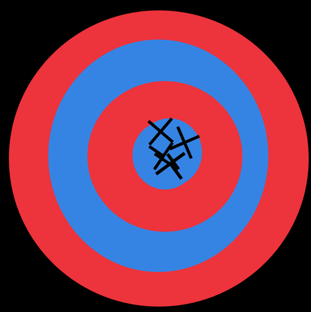
	Accuracy (for multiple results)	**Precision**
Definition	Closeness of the mean of measurements to the true value	Closeness of (repeated) measurements to each other
What affects it	Systematic error	Random error
What is compares	Mean/ vs. true value	Measurements vs. each other
	Accurate: Average of the measurements is close to the true value.	**Precise**:The measurements have a small range.
Note:	Accuracy for a single value relies on both systematic and random errors	



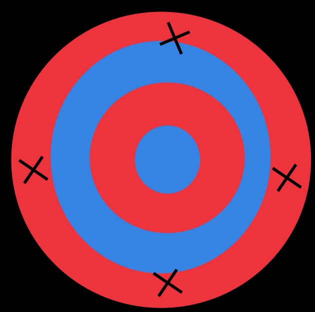
High precision
Low accuracy



Low precision
High accuracy



High precision
High accuracy



Low precision
Low accuracy single vaule
High accuarcy for mean measurements (mutliple)

Uncertainty

Types of Errors

A. Systematic Errors (Bias)

Definition: Consistent, repeatable errors in the same direction

Causes: Faulty calibration, zero error, experimental design flaw, incorrect technique

Effect on results: which makes all readings either higher or lower than **true value**

Examples:

- Ruler with worn zero mark
- Stopwatch that runs slow
- Always measuring from wrong eye position (parallax)

How to identify: Results consistently above/below expected value

How to reduce:

- ✓ Chech for zero error (zero instruments before use)
- ✓ Calibrate instruments regularly
- ✓ Use correct methods/techniques
- ✓ Repeat with different equipment

B. Random Errors (Noise)

Definition: Unpredictable variations in measurements

Causes: Human reaction time, environmental fluctuations, reading variations

Effect on results: Reduces precision (causes scatter)

Examples:

- Slightly different stopwatch reaction times
- Air currents affecting pendulum
- Estimating between scale divisions

How to identify: Scatter in repeated measurements

How to reduce:

- ✓ **Take multiple readings (at different rotations/ positions) and take average**
- ✓ Use **more precise instruments** (to more decimal place/S.F.)

Absolute Uncertainty (Δ)

Absolute Uncertainty (Δ) = The ± value that represents the possible error in a measurement.
Format: Measurement ± Δ (units)

Rounding Rules for Absolute Uncertainty (Δ)

Rule 1: Δ gets 1 Significant Figure

Δ is rounded to 1 significant figure

Rule 2: Measurement (x) matches Δ's decimal places

The measurement is rounded to the same decimal place as Δ

Value	Raw Δ	Final Value
24.356 cm	0.156	24.4 ± 0.2 cm
9.81234 m/s^2	0.0423	9.81 ± 0.04 m/s^2
156.7 g	1.67	157 ± 2 g
3.4567 s	0.00382	3.457 ± 0.004 s

Percentage Uncertainty

Percentage Uncertainty= Uncertainty expressed as a percentage of the measurement.

Formula:
Percentage Uncertainty= (Δ/Value)×100%

Example:
For 24.5±0.2 cm:
%U= (0.2/24.5) × 100% ≈0.82%

Combined Uncertainties

RULE 1: Addition/Subtraction → Add ABSOLUTE Uncertainties

When: $a = b \pm c$

$\Delta a = \Delta b + \Delta c$

RULE 2: Multiplication/Division → Add PERCENTAGE Uncertainties

When: $a = b \times c$ OR $a = \frac{b}{c}$

$\%Ua = \%Ub + \%Uc$

RULE 3: Powers → Multiply Percentage Uncertainty

When: $a = b^n$

$\%Ua = n \times \%Ua$

Examples

① $m_t = 2m_1 + m_2$
$m_1 = 51 \pm 0.7$
$m_2 = 25 \pm 0.7$
– $m_t = (2\times51) + 25 = 127$
– $\Delta m = (2\times0.7) + 0.7 = 2.1$
Result: 127 ± 2

② $E_k = \frac{1}{2}mv^2$
$m = 2.00 \pm 0.01$ kg
$v = 3.0 \pm 0.1$ ms^{-1}

① → $\%U = \frac{0.01}{2} \times 100\% = 0.5\%$

② → $\%U = \frac{0.1}{3} \times 100\% = 3.33\%$

③ $E_k = \frac{1}{2} \times 2 \times 3^2 = 9$ J

④ $\%UE_k = 0.5\% + 2\times3.33 = 7.16\% \rightarrow \Delta E_k = 7.16\% \times 9 = 0.644$

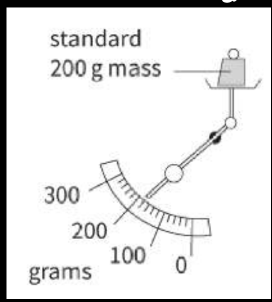
⑤ Result = 9.0 ± 0.6

Finding value of uncertainty

Method 1: Using the Scale Division (Single Measurement)

For Digital Instruments:
Uncertainty = ± smallest displayed unit
Example: Digital thermometer shows 24.5°C → 24.5 ± 0.1°C

For Analog Instruments (rulers, analog meters):
Case 1: You can read inbetween divisions
Uncertainty = ± half of the smallest division
Example: Ruler with mm marks (smallest division = 1 mm)
Measurement: 3.6 cm
Smallest division = 0.1 cm
Uncertainty = ½ × 0.1 = 0.05 cm
Final: 3.60 ± 0.05 cm
Case 2: You cannot read inbetween divisions
Uncertainty = ± smallest division
Example: lever-arm balance (smallest division = 20 g)
Measurement: 180g
Smallest division = 20g
Uncertainty = 20 g
Final: 180 ± 20 g



Method 2: Repeating Readings (Multiple Measurements)

Use this when you can take the same measurement several times

Step-by-Step Procedure:

- Calculate the mean (average)
Mean=Sum of all readings ÷ Number of readings
- Find the range
Range=Maximum reading – Minimum reading
- Calculate uncertainty
Uncertainty=Range ÷ 2
- Report result
Final value=Mean ± Uncertainty

Example: Measuring a Period of Oscillation

Readings: 2.3 s,2.5 s, 2.4 s, 2.6 s, 2.4 s
Mean = (2.3 + 2.5 + 2.4 + 2.6 + 2.4) ÷ 5 = 12.2 ÷ 5 = 2.44 s
Range = Maximum (2.6) – Minimum (2.3) = 0.3 s
Uncertainty = Range ÷ 2 = 0.3 ÷ 2 = 0.15 s

Final result = 2.44 ± 0.15 s ≈ 2.4 ± 0.2 s (rounded to 1 decimal place to match precision)

Apparatus	Typical Uncertainty	Common Use
Meter Rule (1 m)	±1 mm → ±0.5 mm	Measuring length, height, width.
Vernier Calipers / Micrometer Screw Gauge	±0.01 mm	Measuring small thicknesses (wires, sheets).
Analog Balance	±0.1 g → ±0.5 g	Approximate mass measurements.
Graduated Cylinder/ Burette /Beaker	±1 ml → ±0.5 ml	Measuring liquid volumes approximately.
Protractor	1° → ±0.5°	Measuring angles
Others	Follow the general rules mentioned before to determine the uncertainty	
	Note:Remember that sometime an apparatus is used more than one time, for example: to find the length of an object using a meter ruler you would have to read the meter twice (initial and final positions) → 2 x ±0.5 mm = ±1 mm	